

HASSI MESSAoud, Algeria

WMO: 605810

Lat: 31.673N

Lon: 6.140E

Elev: 141

StdP: 99.64

Time Zone: 1.00 (E01)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	2.1	3.7	-7.2	2.1	16.2	-5.7	2.4	15.8	13.9	17.1	12.2	16.8	0.8	200	0.520

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	14.7	46.2	22.2	45.1	21.9	43.8	21.5	24.1	39.2	23.2	39.1	22.6	38.9	5.0	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
19.9	14.9	30.1	18.6	13.7	29.3	17.3	12.6	29.2	72.9	38.5	69.4	38.5	66.8	38.8	

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12.8	10.7	9.3	-0.3	48.2	1.1	1.0	-1.1	48.9	-1.8	49.4	-2.4	50.0	-3.2	50.7
			-1.7	26.8	1.3	2.1	-2.7	28.3	-3.5	29.6	-4.3	30.7	-5.2	32.2
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	23.7	11.7	13.9	18.3	22.9	28.1	32.7	35.7	35.0	30.9	24.8	17.3	12.7	
	DBStd	9.07	2.40	3.30	3.86	3.95	3.90	3.42	2.65	2.37	3.51	3.56	3.24	2.76	
	HDD10.0	20	10	4	0	0	0	0	0	0	0	0	0	6	
	HDD18.3	625	206	130	48	6	0	0	0	0	1	58	176		
	CDD10.0	5031	62	113	259	387	561	682	798	774	627	458	219	91	
	CDD18.3	2594	0	6	49	142	303	432	539	516	377	201	27	2	
	CDH23.3	40838	6	97	626	1794	4371	7196	9612	8921	5657	2269	269	22	
	CDH26.7	27227	0	23	228	889	2658	5000	7158	6469	3622	1113	64	3	
Wind	WSAvg	3.7	2.9	3.6	4.1	4.6	4.7	4.5	3.8	3.6	3.9	3.2	2.8	2.8	
Precipitation	PrecAvg	40	7	2	7	4	4	1	0	1	4	3	4	4	
	PrecMax	117	59	27	47	45	24	3	1	9	33	16	33	22	
	PrecMin	4	1	0	1	0	0	0	0	0	0	0	0	0	
	PrecStd	23	12	5	11	11	5	1	0	2	7	5	7	6	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	24.0	29.3	34.2	39.2	44.0	46.8	48.2	47.2	44.9	39.2	30.9	26.0	
		MCWB	12.7	14.3	16.4	18.7	20.8	21.9	22.5	22.6	21.5	20.4	16.6	14.8	
	2%	DB	21.8	26.1	31.8	36.5	41.1	45.1	47.0	45.8	42.6	36.9	28.5	22.7	
		MCWB	12.2	13.4	15.7	17.9	19.9	21.3	22.5	22.2	21.3	19.2	15.9	13.1	
	5%	DB	20.0	23.8	29.1	34.1	39.0	43.4	45.7	44.1	41.0	34.5	26.4	20.8	
		MCWB	11.6	12.5	14.7	17.1	19.4	20.9	22.2	21.8	21.0	18.9	15.2	12.4	
	10%	DB	18.2	21.2	26.9	31.9	37.0	41.5	44.0	42.8	39.0	32.2	24.2	19.1	
		MCWB	10.6	11.4	14.0	16.5	18.9	20.4	21.8	21.6	20.8	18.5	14.3	11.6	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	14.3	16.5	18.6	20.6	23.2	23.7	25.9	25.4	24.0	22.4	19.1	16.1
			MCDB	20.9	25.4	27.4	33.6	36.3	39.1	41.1	40.0	35.3	33.0	26.9	22.8
		2%	WB	13.0	14.3	16.7	19.0	21.4	22.3	24.0	24.1	23.1	21.1	17.4	14.2
			MCDB	19.7	24.3	27.9	32.1	36.2	39.7	41.2	39.2	35.7	31.1	25.4	20.5
5%		WB	12.0	12.9	15.5	18.1	20.3	21.6	23.0	23.2	22.4	20.3	16.2	13.1	
		MCDB	18.6	22.0	27.3	31.0	35.5	39.2	41.3	39.2	35.4	30.0	24.2	19.4	
10%		WB	11.0	11.8	14.4	17.0	19.4	21.0	22.2	22.6	21.7	19.5	15.0	12.2	
		MCDB	17.4	20.2	25.5	29.6	34.5	38.7	41.1	39.1	35.0	29.3	22.9	18.3	
Mean Daily Temperature Range	5% DB	MDBR	13.3	13.5	13.9	13.9	14.0	14.3	14.7	14.0	13.3	13.0	13.0	12.6	
		MCDBR	15.4	16.2	16.7	16.9	16.0	16.4	16.3	15.7	15.7	15.3	14.7	14.3	
		MCWBR	8.2	7.8	7.3	6.7	6.0	5.9	5.8	5.3	5.3	5.8	6.4	7.5	
	5% WB	MCDBR	14.0	14.5	15.0	14.7	14.4	14.5	14.7	13.6	13.2	12.7	13.0	12.7	
		MCWBR	8.0	7.6	7.2	6.7	6.3	5.9	6.0	5.5	5.0	5.4	6.4	7.2	
Clear-Sky Solar Irradiance	taub	0.371	0.398	0.506	0.550	0.605	0.601	0.619	0.592	0.594	0.519	0.426	0.383		
	taud	2.093	2.008	1.731	1.663	1.574	1.593	1.577	1.636	1.641	1.774	1.956	2.062		
	Ebn at Noon	827	843	773	760	722	722	706	717	690	708	752	786		
	Edh at Noon	130	157	223	248	274	268	271	252	241	196	149	128		
All-Sky Solar Radiation	RadAvg	3.74	4.67	5.72	6.75	7.25	7.73	7.70	7.07	5.94	4.84	3.85	3.32		
	RadStd	0.15	0.20	0.23	0.26	0.25	0.23	0.11	0.15	0.19	0.19	0.14	0.15		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (2 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page